

DECLARATION OF CONFORMITY (EU)

No. 20544/2025

1. Product Name and Model:

Electric Toothbrush Charging Stand / Charging Dock

Model No.: ETCS-3101

2. Importer:

VATSPEEDSL

Calle Antonio Salvador N99.1, 28026 Madrid, Spain

Email: services@vatspeed-eu.com

3. This declaration of conformity is issued under the sole responsibility of the importer:

VATSPEEDSL

Calle Antonio Salvador N99.1, 28026 Madrid, Spain

4. Product Name and Photo:

Electric Toothbrush Charging Stand



5. Product complies with the directives:

- **Low Voltage Directive (LVD) 2014/35/EU**
- **Electromagnetic Compatibility (EMC) Directive 2014/30/EU**
- **RoHS Directive 2011/65/EU – Restriction of Hazardous Substances**
- **REACH Regulation EC No 1907/2006 (chemical safety)**
- **Ecodesign Directive 2009/125/EC – Energy-related Products (standby power consumption)**

- ERP Regulation (EU) 2019/1782 – External power supplies (if charger includes detachable AC/DC adapter)

6. Product complies with the standards:

- EN 60335-1:2012+A15:2021 – Household and similar electrical appliances – Safety – Part 1: General requirements
- EN 60335-2-65:2015 – Particular requirements for air-cleaning appliances (if applicable – not directly, but same inductive charging principles may apply) – Or more specifically: EN 60335-2-29:2021 – Battery chargers
- EN 55014-1:2021 – EMC – Requirements for household appliances – Emission
- EN 55014-2:2021 – EMC – Requirements for household appliances – Immunity
- EN 61000-3-2:2014 – Harmonics
- EN 61000-3-3:2013 – Flicker
- EN 50564:2011 – Electrical and electronic equipment – Measurement of low power consumption (standby/off mode)
- IEC 62321 Series – Determination of six regulated substances (RoHS)
- EN 62479:2010 – Low power electronic equipment – Human exposure to electromagnetic fields (for inductive charging field strength)
- ISO 9001:2015 – Quality management systems

7. Additional Notes:

This electric toothbrush charging stand / charging dock has been tested and found to comply with the applicable EU directives and harmonized standards. It is designed to wirelessly charge/recharge electric toothbrushes using inductive (contactless) charging technology. The product is intended for use in household bathrooms or other indoor environments.

Product meets requirements for:

Electrical Safety (LVD 2014/35/EU):

- Input voltage: 100-240V AC, 50/60 Hz (for models with integrated AC plug)
Or
Input voltage: 5V DC / 12V DC (for USB-powered or low-voltage docking stations)

- Rated power: $\leq 5\text{W}$ (typical for toothbrush chargers)
- Insulation class: Class II (double insulation) or Class III (SELV - Safety Extra Low Voltage)
- Dielectric strength: Withstands 1,500V AC for 1 minute (between primary and secondary windings/accessible parts)
- Leakage current: $< 0.25\text{ mA}$ for Class II appliances
- Protection against electric shock: IPX4 or IPX7 (splash-proof or immersion-proof – typical for bathroom use)
- Overheating protection: Thermal fuse or PTC current limiting in primary circuit
- Short-circuit protection: Output current limited to $\leq 1\text{A}$ for safe operation
- Construction: No accessible live parts when charging stand is open; reinforced insulation between primary (mains) and secondary (charging coil)

Electromagnetic Compatibility (EMC 2014/30/EU):

- Radiated emissions: EN 55014-1 Class B limits (suitable for residential environments)
- Conducted emissions: Meets limits for household appliances
- Electrostatic discharge (ESD): $\pm 8\text{ kV}$ air, $\pm 4\text{ kV}$ contact – no damage or malfunction
- Radiated immunity: 3 V/m , $80\text{ MHz} - 1\text{ GHz}$ – no functional degradation
- Fast transients/burst: $\pm 1\text{ kV}$ on AC mains lines – no upset
- Conducted immunity: 3 V , $150\text{ kHz} - 80\text{ MHz}$ – no upset
- Magnetic field immunity: 3 A/m , $50/60\text{ Hz}$ (power frequency magnetic field)
- Voltage dips/interruptions: Compliant with EN 61000-4-11 – no unsafe state

Human Exposure to Electromagnetic Fields (EMF):

- Inductive charging frequency: $50\text{ kHz} - 500\text{ kHz}$ (typical for toothbrush chargers)
- H-field strength at 10 cm : $< 6.25\text{ A/m}$ (per EN 62479 / EN 50366 / EN 62233 reference levels – ICNIRP 2010)

- Health protection: Meets basic restrictions for general public exposure (no SAR testing required for low-power inductive chargers)

Energy Efficiency (Ecodesign / ERP):

- Standby power consumption: ≤ 0.5 W (no-load condition, when toothbrush not seated)
- Off-mode power consumption: ≤ 0.3 W (if power switch or auto-shutoff circuit present)
- Efficiency: For chargers with detachable power supply, meets EU 2019/1782 minimum efficiency requirements ($>73\%$ for ≤ 5 W adapters)

Materials & Chemical Safety:

- Housing material: ABS / PC / ABS+PC blend – UL94 V-0 or V-2 flame rating
- Surface coating: No toxic heavy metals (lead, cadmium, mercury) – RoHS compliant
- REACH compliance: No SVHC above 0.1% in all plastic and metal components
- Phthalate content: Phthalate-free in PVC parts (if used) – below 0.1% per REACH Annex XVII
- Nickel release: EN 1811:2011 compliant for metal contacts that may be touched (≤ 0.5 $\mu\text{g}/\text{cm}^2/\text{week}$)
- BPA/BPS-free: No bisphenol-A or S in plastic parts in contact with moist environment (optional but recommended for bathroom safety perception)

Mechanical & Environmental Requirements:

- Ingress protection: IPX4 (splash-proof) or IPX7 (immersion-proof up to 1m for 30 minutes)
 - *Note: Many toothbrush charging stands are IPX7 as they may be rinsed under water.*
- Mechanical strength: Housing withstands 0.5 J impact test (steel ball drop or spring hammer)
- Drip-proof design: Drainage holes in base (if sealed) to prevent water accumulation
- Thermal cycling: -10°C to $+40^{\circ}\text{C}$ operating temperature (typical)

indoor/household bathroom range)

- Humidity resistance: 93% RH at 40°C for 48 hours – no condensation ingress
- Cable bending/pull test: If USB cable attached, withstands 2,500 bends (30N pull force) per EN 60335-1

Key Design Specifications:

- Compatible toothbrush models: (list if specific – e.g., Philips Sonicare, Oral-B, generic)
- Charging technology: Inductive / Qi-based (custom coil) – non-radiative near-field magnetic induction
- Typical charging time: 8–24 hours for full toothbrush battery (depends on toothbrush internal battery)
- Output power (to toothbrush): $\leq 1.5\text{W}$ (wireless inductive transfer)
- Base footprint: diameter 50–80 mm, height 20–40 mm
- Weight: 50–200 g (depends on AC plug integrated vs. USB-only version)
- Color options: White, black, pastel (custom)
- Indicator LED: Single LED (red/green/blue) – indicates power-on and charging status (low opto-isolation from mains if present)

Safety Warnings (to be included in product manual/labeling):

- For mains voltage models: Do not immerse charging stand in water – This is a common incorrect belief. Actually, IPX7 models may be submerged briefly, but users should still avoid immersion to be safe. Clarify based on actual IP rating.
- Do not insert foreign objects into charging receptacle.
- Unplug before cleaning (for non-sealed models).
- Use only with supplied power adapter (if detachable type) that matches stand's input rating.
- Do not use if charging stand is damaged or cracked – risk of electric shock.
- Keep away from children under 3 years. Supervise children 3–8 years.
- For bathroom use only if IPX4 or higher (indicated by water drop symbol). Not for use in wet rooms with floor drainage.

Date: 2025-01-01

Place: Madrid, Spain

Authorized Signatory:

Name: Carlos Fernández

Position: Compliance Officer

A handwritten signature in black ink, appearing to read 'Carlos Fernández', with a stylized flourish at the end.

Signature:

A large, bold, black 'CE' logo, likely representing a company or organization, set against a light gray background.